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United States Patent	12402246
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Tagashira; Nobuo

Circuit board

Abstract

A heat radiating substrate includes an insulating layer (11) (insulating substrate) and a circuit pattern (20) of a metal provided on the insulating layer (11) in direct contact with the insulating layer (11), in which a contour line of the circuit pattern (20) has a corner portion (27) that forms a curved line exhibiting a circular arc with a radius of 0.2 mm or more and 5 mm or less when viewed in a top view.

Inventors:	Tagashira; Nobuo (Tokyo, JP)
Applicant:	SUMITOMO BAKELITE CO., LTD. (Tokyo, JP)
Family ID:	1000008779278
Assignee:	SUMITOMO BAKELITE CO., LTD. (Tokyo, JP)
Appl. No.:	18/002711
Filed (or PCT Filed):	June 01, 2021
PCT No.:	PCT/JP2021/020805
PCT Pub. No.:	WO2021/261179
PCT Pub. Date:	December 30, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230247762 A1	Aug. 03, 2023

Foreign Application Priority Data

JP	2020-110347	Jun. 26, 2020
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Publication Classification

Int. Cl.: H05K1/02 (20060101)

U.S. Cl.:

CPC H05K1/0296 (20130101); H05K2201/068 (20130101)

Field of Classification Search

CPC: H05K (1/0296); H05K (2201/068); H05K (2201/09272); H05K (1/056); H05K (1/0209); C04B (2237/366); C04B (2237/368); C04B (2237/407); C04B (37/021); H01L (23/142); H01L (23/49844); H01L (23/3735)

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a circuit board.

BACKGROUND ART

(2) Various developments have been made so far in circuit boards (also referred to as heat radiating substrates) having a heat radiation function. As this kind of technology, for example, a technology disclosed in Patent Document 1 is known. Patent Document 1 discloses a power module in which a semiconductor element is mounted on a support, such as a lead frame, and the support and a heat radiating plate connected to a heat sink are bonded to each other with an insulating resin layer.

RELATED DOCUMENT

Patent Document

(3) Patent Document 1 Japanese Unexamined Patent Publication No. 2011-216619

SUMMARY OF THE INVENTION

Technical Problem

(4) In recent years, better heat radiation properties are required for such a circuit board. Meanwhile, high heat radiation properties are required to be resistant to thermal stress because heat generated by a mounted electronic component or the like increases. That is, it is required that the circuit pattern is not peeled off and is in close contact with the resin substrate even in a case where the above thermal stress acts on the circuit pattern provided on the resin substrate.

(5) The technology disclosed in Patent Document 1 cannot satisfy such a demand, and a new technology has been required.

(6) The present invention has been made in view of such a situation, and an object of the present invention is to provide a technique of improving adhesiveness between a resin substrate and a metal circuit pattern in a case where the metal circuit pattern is provided on the resin substrate in direct contact with the resin substrate.

Solution To Problem

(7) It is possible to provide a circuit board including: an insulating substrate; and a circuit pattern of a metal provided on the insulating substrate in direct contact with the insulating substrate, in which a contour line of the circuit pattern has a corner portion that forms a curved line exhibiting a circular arc with a radius of 0.2 mm or more and 5 mm or less when viewed in a top view.

Advantageous Effects Of Invention

(8) According to the present invention, it is possible to provide a technique of improving adhesiveness between a resin substrate and a metal circuit pattern in a case where the metal circuit pattern is provided on the resin substrate in direct contact with the resin substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a plan view of a heat radiating substrate according to an embodiment.

(2) FIG. 2 is a cross-sectional view of the heat radiating substrate according to the embodiment.

(3) FIG. 3 is an enlarged view of a region A of FIG. 1 according to the embodiment.

(4) FIG. 4 is a chart showing a production step of the heat radiating substrate according to the embodiment.

(5) FIG. 5 (a) and FIG. 5 (b) are diagrams showing cross-sectional structures of circuit patterns of Example and Comparative Example according to the embodiment.

DESCRIPTION OF EMBODIMENTS

(6) Hereinafter, an embodiment of the present invention will be described with reference to the drawings.

(7) <Overview of Heat Radiating Substrate>

(8) FIG. 1 is a plan view of a heat radiating substrate 10. FIG. 2 is a partial cross-sectional view of the heat radiating substrate 10.

(9) The heat radiating substrate 10 is a circuit board on which, for example, an electronic component of a heat-generating body is mounted, is composed of a metal substrate 12, an insulating layer 11 (insulating substrate), and a circuit pattern 20, and is a laminate (laminated body) laminated in this order from the bottom as shown in FIG. 2. The electronic component or the like is mounted on the circuit pattern 20.

(10) A total thickness T0 of the heat radiating substrate 10 is not particularly limited, but is, for example, preferably 300 μm or more and 5000 μm or less and more preferably 1000 μm or more and 4000 μm or less.

(11) <Metal Substrate 12>

(12) The metal substrate 12 is a layer composed of a metal material, and in the present embodiment, the insulating layer 11 is formed on an upper surface thereof, and a heat radiating means (not shown), such as heat radiating fins and radiators, is appropriately attached to the lower surface thereof.

(13) The metal material constituting the metal substrate 12 is not limited to a specific type, but for example, copper, a copper alloy, aluminum, an aluminum alloy, or the like can be used.

(14) A thickness T1 of the metal substrate 12 is not particularly limited, but is the thickest among the elements (the insulating layer 11, the metal substrate 12, and the circuit pattern 20 laminated in the heat radiating substrate 10, and is preferably 10% to 90% with respect to the total thickness T0. The upper limit value of the thickness T1 of the metal substrate 12 is, for example, 20.0 mm or less and preferably 5.0 mm or less. The metal substrate 12 having the thickness T1 equal to or less than the upper limit value is used, whereby the heat radiating substrate 10 as a whole can be made thinner. Further, it is possible to improve the workability in routing, cutting, or the like of the heat radiating substrate 10.

(15) The lower limit value of the thickness T1 of the metal substrate 12 is, for example, 0.1 mm or more, preferably 0.5 mm or more, and more preferably 1.0 mm or more. The metal substrate 12 having the lower limit value or more is used, whereby the heat radiation properties of the heat radiating substrate 10 as a whole can be improved.

(16) <Insulating Layer 11>

(17) The insulating layer 11 is a layer of a resin substrate mainly containing a resin material, and has a function of insulating the metal substrate 12 from the circuit pattern 20. As the insulating layer 11, a ceramic substrate an aluminum nitride substrate, a silicon nitride substrate, or the like) may be used instead of the resin substrate.

(18) The resin material constituting the insulating layer 11 is not limited to a specific type, but examples thereof include a thermosetting resin, such as an epoxy resin, a phenol resin, a urea resin, a melamine resin, a polyester (unsaturated polyester) resin, or a polyimide resin, a silicone resin, and a polyurethane resin. As the resin material, one or a mixture of two or more of these resins can be used.

(19) A filler composed of particles having electrical insulation properties and high thermal conductivity can also be mixed into the resin material constituting the insulating layer 11. Examples of the constituent material of the particles of the filler include metal oxides, such as alumina, and nitrides, such as boron nitride.

(20) A thickness T2 of the insulating layer 11 is appropriately set according to the purpose, but the

thickness T2 of the insulating layer **11** is preferably 40 μm or more and 400 μm or less from the viewpoint of more effectively transferring heat from the electronic component to the metal substrate **12** while improving the mechanical strength and heat resistance, and the thickness T2 of the insulating layer **11** is more preferably set to 80 μm or more and 300 μm or less from the viewpoint of further excellent balance between the heat radiation properties and the insulation properties in the entire heat radiating substrate **10**. The thickness T2 of the insulating layer **11** is set to the above upper limit value or less, whereby it is possible to facilitate the transfer of heat from the electronic component to the metal substrate **12**. In addition, the thickness T2 of the insulating layer **11** is set to the above lower limit value or more, whereby the insulating layer **11** can sufficiently alleviate the generation of thermal stress caused by a difference in coefficient of thermal expansion between the metal substrate **12** and the insulating layer **11**. Further, the insulation properties of the heat radiating substrate **10** are improved.

(21) The physical properties of the resin material forming the insulating layer **11** will be described.

(22) The lower limit value of the glass transition temperature Tg of the resin material is, for example, 100° C. or higher, preferably 125° C. or higher, and more preferably 150° C. or higher. With this, the thermal decomposition of the cured product of the resin material (thermosetting resin composition) can be restrained. Further, the warpage of the insulating layer **11** (that is, the heat radiating substrate **10**) can be restrained. As a result, it is possible to restrain the positional deviation of the electronic component or the like mounted on the circuit pattern **20** with respect to the heat radiating substrate **10**, and it is possible to further enhance the connection reliability between the electronic component or the like and the heat radiating substrate **10**. On the other hand, the upper limit value of the glass transition temperature Tg is not particularly limited, but may be, for example, 300° C. or less. The glass transition temperature of the insulating layer **11** can be measured on the basis of JIS C 6481.

(23) The insulating layer **11** has high thermal conductivity. Specifically, the thermal conductivity of an insulating resin layer **102** in the thickness direction, which is measured by a laser flash method, is preferably 3.0 W/m.K or more and more preferably 5.0 W/m.K or more, with this, heat from the electronic component or the like mounted on the circuit pattern **20** can be easily transferred to the metal substrate **12** through the insulating layer **11**.

(24) An elastic modulus E of the insulating layer **11** at 25° C. is, for example, 1 GPa or more and 50 GPa or less. In addition, the lower limit value is preferably 5 GPa or more and more preferably 10 GPa or more from the viewpoint that the positional deviation of the electronic component with respect to the circuit board can be restrained as a result of increasing stiffness and reducing the warpage of the insulating resin layer **102**, and the connection reliability between the electronic component and the circuit board can be further enhanced. The upper limit value is not particularly limited, but is 50 GPa or less as a realistic value thereof, as described above.

(25) The above elastic modulus E can be measured with a dynamic viscoelasticity measuring device, and is a value of the elastic modulus at 25° C., for example, when measured at a temperature from -5° C. to 300° C., a frequency of 1 Hz, and a heating rate of 5° C. to 10° C./min with a tensile load applied to the insulating layer **11**.

(26) <Circuit Pattern **20**>

(27) The circuit pattern **20** is composed of a conductive metal material, and is electrically connected to an electronic component (LED or the like) of a heat-generating body by, for example, soldering. For example, copper can be suitably used as the metal material constituting the circuit pattern **20**. With this, the circuit pattern **20** has a relatively small resistance value. At least a part of the circuit pattern **20** may be covered with a solder resist layer.

(28) The circuit pattern **20** is formed, for example, by working a metal layer laminated on an insulating layer upper surface of the insulating layer **11** into a predetermined pattern through cutting and etching. The formation process will be described later in FIG. 4, but in the present embodiment, rolled copper is used as a metal layer **20A** of FIG. 4.

(29) The lower limit value of a thickness T3 of the circuit pattern **20** is, for example, 0.3 mm or more. In a case where the lower limit value is such a numerical value or more, it is possible to restrain the circuit pattern **20** from generating heat even in a use requiring a high current. Further, the upper limit value of the thickness of the circuit pattern **20** is, for example, 5.0 mm or less, preferably 4.0 mm or less, and more preferably 3.0 mm or less, in a case where the upper limit value is such a numerical value, the circuit workability can be improved, and the heat radiating substrate **10** as a whole can be made thinner.

(30) <Detailed Shape of Circuit Pattern **20**>

(31) FIG. **3** is an enlarged view of a region A of FIG. **1**, and specifically shows an upper right corner portion **27** of the circuit pattern **20** of the heat radiating substrate **10**.

(32) A contour line of the circuit pattern **20** has the corner portion **27** that forms a curved line exhibiting a circular arc with a radius of 0.2 mm or more and 5 mm or less when viewed in a top view. That is, the circular arc of the corner portion **27** is a circular arc-shaped contour line that interconnects a first pattern contour line **25** extending horizontally in the drawing and a second pattern contour line **26** extending vertically in the drawing. In the present embodiment, the upper right corner portion **27** is shown as a curved line exhibiting the circular arc with the above radius, but four corner portions of the circuit pattern **20** are each preferably a curved line exhibiting the circular arc with the above radius.

(33) The lower limit value of the radius of the circular arc exhibited by the corner portion **27** is preferably 0.5 mm or more. The upper limit value of the radius is not particularly limited, but is preferably 4 mm or less as a realistic value for mounting.

(34) <Interface Between circuit Pattern **20** and insulating Layer **11**>

(35) The structure of the interface between the circuit pattern **20** and the insulating layer **11** in a case where the insulating layer **11** consists of a resin and the metal of the circuit pattern **20** consists of rolled copper will be described by focusing on the thermal stress generated therefrom.

(36) The linear expansion coefficient at a temperature equal to or lower than the glass transition temperature of the resin of the insulating layer **11** is denoted by $\alpha_{\text{sub.R}}$. The linear expansion coefficient $\alpha_{\text{sub.R}}$ is, for example, 10 to 50 ppm/K. The thermal expansion coefficient can be measured using a linear expansion coefficient measuring method conforming to JIS Z2285:2003.

(37) The linear expansion coefficient of the rolled copper of the circuit pattern **20** is denoted by $\alpha_{\text{sub.Cu}}$. $\alpha_{\text{sub.Cu}}$ is about 17 ppm/K.

(38) The glass transition temperature of the resin of the insulating layer **11** is denoted by $T_{\text{sub.g}}$, and the elastic modulus thereof is denoted by E. The glass transition temperature $T_{\text{sub.g}}$ is, for example, 100° C. to 300° C. as described above. The elastic modulus E is, for example, 5 to 50 GPa.

(39) In a case where the room temperature at the time of measurement is denoted by R, a stress index S calculated by Equation (1) is 1 MPa or more and 50 MPa or less. The stress index S indicates the stress factor when the rolled copper is provided on the resin, that is, the stress generated at the interface between the insulating layer **11** and the circuit pattern **20**, and is preferably low. The upper limit value is more preferably 25 MPa or less and still more preferably 20 MPa or less. Although the lower limit value is not particularly limited, appropriate adhesiveness can be achieved even at 1 MPa or more, in a case where the contour line of the corner portion **27** exhibits a circular arc with a radius of 0.2 mm or more and 5 mm or less when viewed in a top view. In particular, in a case where the circuit pattern **20** is formed by etching after cutting with a router, as will be described in the following production process, the side surface part of the circuit pattern **20** has a small trailing-tailed shape at the interface part between the insulating layer **11** and the circuit pattern **20**, and interfacial peeling does not easily occur. In particular, the desired interface adhesiveness can be achieved even in a case where the stress index S is 1 MPa or more.

$$S = |\alpha_{\text{sub.R}} - \alpha_{\text{sub.Cu}}| \times (T_{\text{sub.g}} - R_{\text{sub.T}}) \times E \quad \text{Equation (1)}$$

<Method of Producing Heat Radiating substrate **10**>

(40) FIG. 4 is a chart showing the production process of the heat radiating substrate **10**.

(41) (S10: Laminate Producing Step (Laminated Body Preparing Step))

(42) A laminate **10A** in which the metal substrate **12**, the insulating layer **11**, and the metal layer **20A** are laminated in this order from the bottom is prepared. The metal layer **20A** is worked by the following steps to form the circuit pattern **20**.

(43) As the method of producing the laminate **10A**, a known method can be used. For example, the metal substrate **12** is used as a carrier, and a liquid material (varnish-like material) as a constituent material of the insulating layer **11** is applied onto the metal substrate **12** having the thickness **T1** by, for example, a spray method.

(44) Then, the liquid material applied on the metal substrate **12** is dried by natural drying or forced drying. With this, the insulating layer **11** having the thickness **T2** can be obtained. At this time, the insulating layer **11** may not be completely cured (so-called B stage state).

(45) Next, the metal layer **20A** having a thickness **T3'** is formed on the insulating layer **11**. That is, the metal layer **20A** (here, rolled copper) to be the circuit pattern **20** is laminated on the insulating layer upper surface of the insulating layer **11** by a hot press or the like. With this, the laminate **10A** can be obtained. The thickness **T3'** of the metal layer **20A** is set in consideration of an etching step, which will be described later.

(46) (S12: Circuit Pattern Cutting Step)

(47) Subsequently, the metal layer **20A** of the above-described laminate **10A** is cut so as to have a desired pattern by using a router. A provisional circuit pattern **20B** is formed on the insulating layer **11** by leaving a metal layer (thin copper portion **20B1**) having a predetermined thickness for a portion that is not a pattern. That is, there is a concern that the insulating layer **11** may be damaged in a case where all the patterns are formed by cutting. Therefore, the metal layer (thin copper portion **20B1**) having the thickness is partially left as a margin. With this, a laminate **10B** having the provisional circuit pattern **20B** can be obtained. At this time, the corner portion **27** is also cut into a slightly larger circular arc in consideration of the next etching step (S14).

(48) (S14: Etching Step)

(49) Subsequently, the etching process is performed on the laminate **10B** having the provisional circuit pattern **20B** to melt the remaining metal layer (thin copper portion **20B1**) and to form a desired pattern so that the final circuit pattern **20** can be obtained. With this, the heat radiating substrate **10** can be obtained. At this time, the corner portion **27** forms a circular arc with a radius of 0.2 mm or more and 5 mm or less when viewed in a top view. With this, the heat radiating substrate **10** can be obtained.

(50) <Effect of Embodiment>

(51) The features and effects of the embodiment are summarized as follows.

(52) (1) The heat radiating substrate **10** includes: the insulating layer **11** (insulating substrate); and the circuit pattern **20** of a metal provided on the insulating layer **11** in direct contact with the insulating layer **11**, in which the contour line of the circuit pattern **20** has the corner portion **27** that forms a curved line exhibiting a circular arc with a radius of 0.2 mm or more and 5 mm or less when viewed in a top view. With this, the adhesiveness of the corner portion **27** can be improved.

(53) (2) In the heat radiating substrate **10**, the insulating layer **11** consists of a resin substrate.

(54) (3) In the heat radiating substrate **10**, the metal of the circuit pattern **20** consists of rolled copper.

(55) (4) In the heat radiating substrate **10**, the insulating layer **11** consists of a resin, and the metal of the circuit pattern **20** consists of rolled copper, and in a case where the linear expansion coefficient at a temperature equal to or lower than the glass transition temperature of the resin is denoted by $\alpha_{\text{sub.R}}$, the linear expansion coefficient of the rolled copper (that is, the circuit pattern **20**) is denoted by $\alpha_{\text{sub.Cu}}$, the glass transition temperature of the resin is denoted by $T_{\text{sub.g}}$, the elastic modulus of the resin is denoted by E , and the room temperature at the time of measurement is denoted by $R_{\text{sub.T}}$, a stress index S calculated by Equation (1) is 1 MPa or more and 50 MPa or

less.

$$S = |\alpha_{\text{sub.R}} - \alpha_{\text{sub.Cu}}| \times (T_{\text{sub.g}} - R_{\text{sub.T}}) \times E \quad \text{Equation (1)}$$

(56) As described above, even in a case where the stress index S is 1 MPa or more and 50 MPa or less, good adhesiveness between the insulating layer **11** and the circuit pattern **20** can be realized.

(57) Although the embodiment of the present invention has been described above, the embodiment is an example of the present invention, and various configurations other than the above embodiment can be adopted.

EXAMPLES

(58) FIG. 5 (a) and FIG. 5 (b) show photographs of cross-sectional structures of Example and Comparative Example. FIG. 5 (a) is the heat radiating substrate **10** produced by forming the circuit pattern **20** by using the cutting and etching shown in the above-described embodiment (Example). FIG. 5 (b) is a heat radiating substrate produced by forming the circuit pattern **20** by using only conventional general etching (Comparative Example). Here, the photographs of the circuit cross-sections are vertically arranged so as to be comparable with each other. In these photographs, the width of the circuit pattern **20**, more specifically, the width of a metal layer lower surface **22** (the interface with the insulating layer **11**) is formed to be 1 mm.

(59) In Comparative Example shown in FIG. 5(b), the side surface portion has a trailing-tailed shape (substantially Mt. Fuji shape) as a whole, and the area of the upper surface of the circuit pattern **20** is narrowed. On the other hand, in Example shown in FIG. 5(a), the region of the side surface portion with the trailing-tailed shape is small, and most of the region forms a straight line. In general, since a resin has a higher coefficient of linear expansion than that of a metal, thermal stress caused by the difference in coefficient of linear expansion between the resin and the metal increases in an environment involving temperature changes. As a result, interfacial peeling between the circuit pattern **20** and the insulating layer **11** may occur. In particular, the interfacial peeling is likely to occur in a region such as the corner portion **27**. However, as shown in the present example, the side surface part of the circuit pattern **20** has a small trailing-tailed shape at the interface part between the insulating layer **11** and the circuit pattern **20**, and the interfacial peeling does not easily occur. In particular, in a case where the stress index S described above is satisfied, the adhesiveness of the interface can be greatly improved, and the interfacial peeling can be restrained, as compared to the conventional structure.

(60) This application claims priority based on Japanese Patent Application No. 2020-110347 filed on Jun. 26, 2020, all of its disclosures are incorporated herein.

REFERENCE SIGNS LIST

(61) **10** heat radiating substrate **10A**, **10B** laminate **10B** laminate **11** insulating layer **12** metal substrate **20** circuit pattern **20A** metal layer **20B** provisional circuit pattern **20B1** thin copper portion **27** corner portion

Claims

1. A circuit board comprising: an insulating substrate; and a circuit pattern of a metal provided on the insulating substrate in direct contact with the insulating substrate, wherein a contour line of the circuit pattern has a corner portion that forms a curved line exhibiting a circular arc with a radius of 0.2 mm or more and 5 mm or less when viewed in a top view, wherein the insulating substrate consists of a resin substrate, wherein the thickness of the circuit pattern is between 0.3 mm and 5 mm, wherein the insulating substrate consists of a resin, and the metal consists of rolled copper, and wherein in a case where a linear expansion coefficient at a temperature equal to or lower than a glass transition temperature of the resin is denoted by α_R , a linear expansion coefficient of the rolled copper is denoted by α_{Cu} , the glass transition temperature of the resin is denoted by T_g , an elastic modulus of the resin is denoted by E , and a room temperature at the time of measurement is denoted by R_T , a stress index S calculated by Equation (1) is 1 MPa or more and 10 MPa or less,

$S = |\alpha_{\text{sub.R}} - \alpha_{\text{sub.Cu}}| \times (T_{\text{sub.g}} - R_{\text{sub.T}}) \times E$ Equation (1).

2. The circuit board according to claim 1, wherein the metal consists of rolled copper.
 3. The circuit board according to claim 1, wherein the elastic modulus of the resin is 1 GPa or more and 50 GPa or less at 25° C.
 4. The circuit board according to claim 1, wherein the glass transition temperature of the resin is between 100° C. and 300° C.
 5. The circuit board according to claim 1, wherein the resin includes at least one or more selected from the group consisting of epoxy resin, phenol resin, urea resin, melamine resin, polyester (unsaturated polyester) resin, polyimide resin, silicone resin and polyurethane resin.
 6. The circuit board according to claim 1, wherein the thickness of the insulating substrate is between 40 μm and 400 μm.
 7. The circuit board according to claim 1, wherein the insulating substrate has a metal substrate on the opposite side of the circuit pattern.
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